Dharmenthiran T

Software Developer

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Summary

Software Developer specializing in building scalable Web, Desktop, MacOS, iOS, and Android applications with a focus on Industrial IoT and ERP architecture. Expert in real-time telemetry, hardware-software integration, and performance optimization for ARM-based edge environments.

Education

Govt Arts And Science College (Autonomous) Coimbatore, TN
Bachelor of Science in Information Technology • 72% *2020 – 2023*

Chirumalar Higher Secondary School Trichy, TN
Higher Secondary (HSC) • 74% *2020*

Experience

Software Development Engineer I Jan 2024 – Present
SPI Equipments India Pvt. Ltd. *Coimbatore, TN*

- Architected cross-platform ERP and industrial automation solutions for Windows, Android, and embedded ARM environments.
- Developed real-time telemetry systems using **FastAPI and WebSockets**, enabling sub-second synchronization for machine control.
- Integrated hardware protocols (TCP/IP, RS232/485) and containerized microservices using **Docker** for scalable deployments.

Technical Skills

Backend: Python (FastAPI, Django, Flask), Dart, SQLAlchemy, Alembic
Frontend: Flutter, Dart, HTML, CSS, Kivy, KivyMD
Databases: SQL, MySQL, PostgreSQL, MongoDB
DevOps & Security: Docker, GitHub Actions, JWT
Connectivity: REST APIs, WebSockets, MQTT, Redis, TCP/IP, Serial Communication (RS232/RS485)
Tools: Git, GitHub, PyCharm, Android Studio, VS Code

Projects

Industrial IoT & Automation | *Flutter, Dart, SQL, Serial Communication, TCP/IP*

- Built a Flutter control system for real-time telemetry and automated dosing using MVVM architecture.
- Integrated batch management and ERP connectivity to optimize production and inventory transparency.

Enterprise Lifecycle Manager (ERP/MES) | *Flutter, FastAPI, MySQL, Docker, JWT*

- Engineered an ERP ecosystem with **Alembic** migrations and secure **RBAC/JWT** access control.
- Implemented high-performance state management and audit logging for mission-critical workflows.

SolarOps – Industrial IoT Monitoring | *Flutter, FastAPI, MQTT, MySQL, Docker, Redis, WebSockets*

- Architected a full-stack IoT ecosystem for real-time monitor and control of industrial solar cleaning machines.
- Integrated **MQTT** to process live telemetry from Raspberry Pi devices with sub-second latency.

ServiceLink (Enterprise SaaS ERP) | *Flutter, FastAPI, MySQL, WebSockets*

- Developed a SaaS platform to automate the service lifecycle from complaint logging to automated billing.
- Engineered real-time syncing and chat using **WebSockets**, significantly reducing coordination latency.